



P000340SE

EVALUATION REPORT #1: TTRACT

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ABSTRACT

This evaluation assessed AWS Textract's `AnalyzeExpense` API for extracting key receipt data, including vendor, date, total, and line items, for integration into ReceiptCAT. Using 118 receipts of varied formats and conditions, the analysis provided a qualitative view of Textract's strengths and limitations. While totals were consistently detected, vendor, line-item, and date data showed irregularities influenced by image quality and layout. The findings, though non-quantitative, informed key enhancements to the `receiptExtractor` Lambda function in the ReceiptCAT application, including the introduction of post-processing and validation logic to improve reliability and consistency.

1 INTRODUCTION

ReceiptCAT is a receipt-item categorisation application that relies on accurate data extraction as the foundation for downstream classification and reporting. Before categorisation can occur, the system must reliably extract key expense details such as vendor names, total amounts, and dates from receipt images.

To evaluate this extraction stage, we assessed AWS Textract's `AnalyzeExpense` API, which converts receipt images into structured JSON output. The goal was to determine how effectively Textract could detect and extract the essential expense fields across a diverse sample of Australian receipts and to identify recurring extraction errors or inconsistencies.

We analysed Textract's performance across multiple receipt formats and vendors, focusing on its field detection accuracy, consistency, and completeness. These findings were used to highlight strengths, limitations, and areas for potential improvement before integrating Textract into the end-to-end ReceiptCAT workflow.

2 EVALUATION PLATFORM AND API

AWS Textract is a managed document analysis service that uses machine learning to extract text and key-value pairs from scanned documents [1]. For expense data, the `AnalyzeExpense` API provides a pre-trained model optimised for receipts and invoices [2].

For this evaluation, Textract was tested in isolation, independent of other ReceiptCAT components. Each receipt was processed using Textract's synchronous `AnalyzeExpenseCommand` API [3], which returned `ExpenseDocuments` containing `SummaryFields` and `LineItemGroups`. `SummaryFields` would contain the vendor name, date and total, and `LineItemGroups` would contain an array of purchase items, where each array element would contain the name, price and quantity of the purchase item.

The scope and boundaries of the Textract evaluation are as follows:

1. It focused solely on Textract's accuracy in extracting data from receipts.
2. It is limited to receipts in JPG and PNG formats, taken with a camera or via screenshots from mobile phone apps (both Android and iOS).
3. It excluded evaluation of non-expense Textract APIs such as `AnalyzeDocument`.
4. Ground-truth precision, recall, and F1 measurements were not performed.

3 METHODOLOGY

3.1 DATASET COLLECTION AND COMPOSITION

The evaluation used 118 receipts collected from Australian vendors. The dataset was manually gathered by the ReceiptCAT development team, consisting of five members. A total of 77 receipts (65%) were physical receipts photographed using three different smartphones: Samsung Galaxy A52 (Android 14),

Google Pixel 7 Pro (Android 16), and iPhone 13 Pro (iOS 18.6.2). The remaining 41 receipts (35%) were electronic receipts obtained as screenshots of digital copies. All 118 receipts were stored as image files (e.g., .jpg, .png), and no PDF-based receipts were included.

Due to time constraints, no image stratification was applied during data collection. The evaluation team did not modify or standardise the photographs to control for factors such as lighting, clarity, or angle. As a result, the dataset contains a natural mix of image conditions rather than predefined experimental categories. This approach allowed the team to focus on testing Textract’s performance on realistic user-generated receipts, with controlled analysis of image quality and receipt condition identified as a future evaluation task.

The characteristics of the collected receipts, including the proportion of physical and electronic copies, the devices used to capture the photos, and the observed physical conditions of the receipts, are summarised in Table 3.1.1.

Attribute	Category	Receipt Count	Percentage (%)
Receipt Format	Physical	77	65.25%
	Electronic (Screenshots)	41	34.74%
Phone Model (for Physical receipts only)	Samsung Galaxy A52 (Android 14)	33	27.97%
	Google Pixel 7 Pro (Android 16)	26	22.03%
	iPhone 13 Pro (iOS 18.6.2)	18	15.26%
Image Quality (for Physical receipts only)	Clear	62	52.54%
	Blurred	15	12.71%
Receipt Condition (for Physical receipts only)	Folded	35	29.66%
	Crumpled	24	20.34%
	Flat	18	15.25%
Angle of Receipt in Image (for Physical receipts only)	Levelled	25	21.19%
	Skewed	52	44.07%

Table 3.1.1 – Overview of receipt composition, capture devices, and physical image characteristics.

The dataset represents 59 distinct vendors, with receipts sourced from a mix of large retail chains, local stores, and service-oriented businesses. This composition mirrors the team's typical spending patterns, with major retailers accounting for roughly one-third of the dataset and the remainder comprising smaller retail and service vendors. The largest contributors were Woolworths and Woolworths Metro, which together accounted for 24% of all receipts. Most other vendors contributed one to two receipts each, spanning industries such as groceries, dining, beauty, electronics, pharmacy, utilities, and professional services. The distribution of receipts across vendors with more than 1 receipt is presented in Table 3.1.2.

Vendor	Receipt Count	Percentage (%)
Woolworths	19	16.10%
Woolworths Metro	10	8.47%
Coles	7	5.93%
Medical Hub RMIT	6	5.08%
BIGW	5	4.24%
Chemist Warehouse	4	3.39%
Sephora	4	3.39%
IGA	4	3.39%
Aldi	4	3.39%
Halaya	3	2.54%
Aussie Broadband	3	2.54%
Joy Market	2	1.69%

Table 3.1.2 – Distribution of vendors across the dataset where the receipt count is more than 1.

To better understand the diversity of receipt sources, all vendors were classified into predefined business categories. The breakdown of receipts across these categories is shown in Table 3.1.3 above.

Vendor Type	Receipt Count	Percentage (%)
Groceries	47	39.83%
Eating Out & Takeaway	30	25.42%
Professional Services	7	5.93%
Personal Care & Beauty	6	5.08%
Department Store	5	4.24%
Pharmacy	5	4.24%
Utilities & Bills	5	4.24%
Stationery & Office	3	2.54%
Clothing & Footwear	2	1.69%
Electronics & Tech	2	1.69%
Gifts & Occasions	2	1.69%
Home & Cleaning	2	1.69%
Sports & Fitness	1	0.85%
Travel & Holidays	1	0.85%

Table 3.1.3 – Categorisation of receipts by vendor type.

The largest vendor type in the dataset was *Groceries*, representing 40% of all receipts, followed by *Eating Out & Takeaway* at 25%. Together, these two categories accounted for approximately two-thirds of all receipts collected.

3.2 DATA EXTRACTION PROCESS

Each receipt was uploaded to an Amazon S3 bucket configured with an event notification that invoked the receipt extraction Lambda function. The Lambda called Textract's *AnalyzeExpense* API for each new object, parsed the response, and wrote the extracted fields to a DynamoDB table. After all receipts had completed processing, the team manually exported the DynamoDB records to a CSV file for detailed analysis. This CSV export was performed only for the evaluation and is not part of the ReceiptCAT production workflow.

Each record in the CSV included:

1. Receipt ID (UUID generated in Lambda)
2. User ID
3. Vendor (extracted from SummaryFields of the AnalyzeExpense API response)
4. Date (extracted from SummaryFields of the AnalyzeExpense API response)
5. Total (extracted from SummaryFields of the AnalyzeExpense API response)
6. Items (list of item name, price and quantity extracted from LineItemGroups of the AnalyzeExpense API response)
7. S3_Path: File path of the receipt uploaded to the S3 bucket
8. Processed_At: Timestamp of receipt processing completion, which is recorded as a Unix epoch timestamp in milliseconds (i.e., the number of milliseconds elapsed since 1 January 1970, 00:00:00 UTC).

The extracted CSV data were manually reviewed by two members (Sneha Thankam Oommen and Gabby Sanchez) of the ReceiptCAT development team. Each reviewer independently analysed a portion of the dataset, comparing the extracted fields with the original receipt images and noting recurring issues or anomalies. The review focused on identifying common extraction patterns related to field presence, image quality, text layout, and language variation.

3.3 CODE AND DATA AVAILABILITY

To ensure reproducibility of this evaluation, all scripts, data, and supporting materials are provided in a zip archive (P000340SE-EvaluationReport-Appendix.zip) accompanying this report. The archive contains both the code and datasets used to reproduce the evaluation, as well as the manually reviewed ground-truth analysis results.

1. **textextract-evaluation-infra** - Contains the AWS CDK application and Lambda code used to deploy the evaluation stack for Textextract. Deploying this stack provisions the S3 bucket, the S3-to-Lambda event trigger, and the DynamoDB table, with all required permissions.
2. **data/textextract_output_data.csv** - Contains the extracted data generated during the evaluation.
3. **textextract_input_receipt_images.zip** - ZIP file containing the input receipt images used in this evaluation.
4. **data/receipt-ground-truth-manual-review.csv** – Row-level annotations from the manual ground-truth review. For each receipt, it records the receipt format (physical or electronic), capture device and photo conditions (image quality, condition, and angle), issuing vendor and vendor type, vendor name printed on the receipt and its format (logo or text), Textextract’s extracted vendor and date along with match indicators against the ground truth, item-extraction presence and counts, and the language used on the receipt.
5. **textextract-analysis_appendix.pdf** - Contains visual examples of receipts and extracted outputs referenced in the Findings section.
6. **README_Textextract_Evaluation.md** - Provides step-by-step deployment and reproduction instructions for running the CDK stack, uploading receipts, and exporting processed data.

4 FINDINGS

Each of the findings below is supported by examples in the **textextract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive, which includes both receipt images and extracted outputs.

The dataset of 118 receipts was manually reviewed to enable quantitative insights from the recorded observations. The manual ground-truth analysis compared each receipt image with its extracted Textextract

output to identify recurring extraction behaviours and issues, such as missing vendor names, inconsistent date formats, or incomplete item lists.

4.1 VENDOR NAME VARIABILITY

Receipts where the printed vendor name was recorded as “Unknown” (12 of 118) were excluded from accuracy calculations because no ground truth vendor text was available for comparison. Metrics below are computed over the remaining 106 receipts.

Textract returned multiple string variants for the same vendor (e.g., “WOOLWORTHS GROUP LIMITED”, “Woolworths”, “Woolworths Online”) (Refer to Findings 1 of the Appendix in **textract-analysis_appendix.pdf** for sample receipts and extracted outputs). These variations arise from differences in layout across stores and receipt types.

Textract achieved similar accuracy for both **text-based (68.7%)** and **logo-based (66.7%)** receipts as shown in Table 4.1.1. Logo-based receipts, however, were slightly more prone to inclusion of additional characters due to line breaks and partial logo segmentation. (Refer to Findings 2 of the Appendix for sample receipts and extracted outputs.)

Format of Vendor Name in Receipt	Total Receipts	Exact Matches	Mismatches	Extracted as “Unknown”	Accuracy (%)
Text	67	46	19	2	68.7%
Logo	39	26	13	0	66.7%

Table 4.1.1 – Performance of Textract across receipts depicting vendor names as text or logo.

4.2 DATE FORMAT INCONSISTENCY

Dates were detected consistently but appeared in varying formats such as 16 Apr 2025, 07/04/25, and 2025-09-12 (Refer to Finding 3 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for date extraction samples.) Without post-processing, these inconsistencies affect sorting and temporal grouping.

4.3 VENDOR MISIDENTIFICATION

When vendor names were absent, Textract sometimes misclassified nearby text as vendor names, including personal identifiers. If no recognisable vendor text existed, “Unknown” was returned. (Refer to Finding 4 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for vendor misidentification examples.)

4.4 LINE-ITEM AND PRICE FORMATTING

Line items displayed irregularities such as missing currency symbols, inconsistent casing, and embedded special characters. Composite or combo products were represented as long strings. When receipts had clear table layouts, line items were captured correctly. (Refer to Finding 5 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for item formatting variations.)

4.5 LANGUAGE LIMITATIONS

Receipts containing non-English product names were partially parsed. Unsupported scripts led to missing item data, confirming that Textract’s model primarily targets English text. (Refer to Finding 6 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for language-related extraction examples.)

4.6 IMAGE QUALITY SENSITIVITY

Image resolution and clarity directly affected extraction completeness. Blurry, angled, or shadowed receipts often resulted in missing or partially extracted fields. Textract also occasionally omitted data from otherwise legible images, indicating sensitivity to lighting and image noise.

However, the current dataset was limited and did not include receipts captured under systematically varied lighting conditions or across multiple versions of the same receipt (e.g., crisp versus wrinkled or folded copies). Only a few examples of this behaviour were observed, and further analysis with a broader dataset is required in the future quantitative evaluation to validate this finding. (Refer to Finding 7 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for examples of extraction issues due to image clarity.)

4.7 INVOICE-STYLE RECEIPTS

Receipts structured as invoices lacked line-item data. Textract still extracted summary fields (vendor, total, date) but left item arrays empty, aligning with its design focus on expense-style layouts. (Refer to Finding 8 of the **textract-analysis_appendix.pdf** file within the P000340SE-EvaluationReport-Appendix.zip archive for invoice-style receipts.)

5 LIMITATIONS

This evaluation was designed as an initial qualitative assessment based on manual review rather than a quantitative, metric-driven analysis. The dataset was collected internally by the development team and therefore lacks the diversity and structured sampling needed for statistical generalisation across vendors, receipt types, and image conditions. Precision, recall, and F1 scores were not computed because the analysis focused on observing recurring extraction issues through direct comparisons between images and outputs. Additionally, the evaluation did not include controlled testing across variables such as lighting, angle, or receipt condition, and only the synchronous AnalyzeExpense API was tested rather than alternative Textract endpoints.

Even with these limitations, the findings were valuable in guiding implementation decisions for ReceiptCAT. They revealed the need for post-processing steps to normalise extracted data (such as date formats and total values) and highlighted cases in which Textract inconsistently detected vendor names, dates, or line items. These insights directly informed improvements to the receiptExtractor Lambda function, ensuring that extracted outputs are standardised and resilient before being passed to the downstream categorisation stage.

6 CONCLUSION AND RECOMMENDATIONS

This evaluation provided an initial, qualitative understanding of AWS Textract's capabilities when applied to receipt images of varied formats and conditions. Textract consistently identified totals across most receipts, demonstrating its potential to extract key expense information without requiring custom model training. However, the evaluation also revealed recurring challenges, including inconsistent vendor detection and missing or irregular line-item and date data.

Although quantitative accuracy metrics were not computed, the findings offered valuable insight into how Textract behaves under real-world conditions. These insights directly informed improvements to ReceiptCAT's OCR pipeline, particularly the receiptExtractor Lambda function. The evaluation demonstrated that while Textract's raw outputs are not immediately production-ready, they can form a solid foundation when complemented with appropriate post-processing and validation logic.

Recommended Actions for the receiptExtractor Function:

1. Apply post-processing logic to standardise extracted dates and price fields. This includes converting dates to a consistent format (YYYY-MM-DD) and ensuring total price values are properly formatted as numeric data.
2. When required fields such as vendor, date, or total are missing, populate them with default values before passing the data downstream. This approach ensures continuity of processing while maintaining placeholders for later correction using the Edit feature in ReceiptCAT.

These recommendations directly informed improvements to the receiptExtractor Lambda function in the ReceiptCAT backend stack. The Lambda function was updated to include date and total price normalisation and to insert default values for missing fields, ensuring that AWS Textract outputs remain consistent and usable for subsequent processing stages.

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